

What Is Deep Learning?

DSC 330 Applied Machine Learning

April 18, 2025

Unit: Deep Learning #1

The Plan

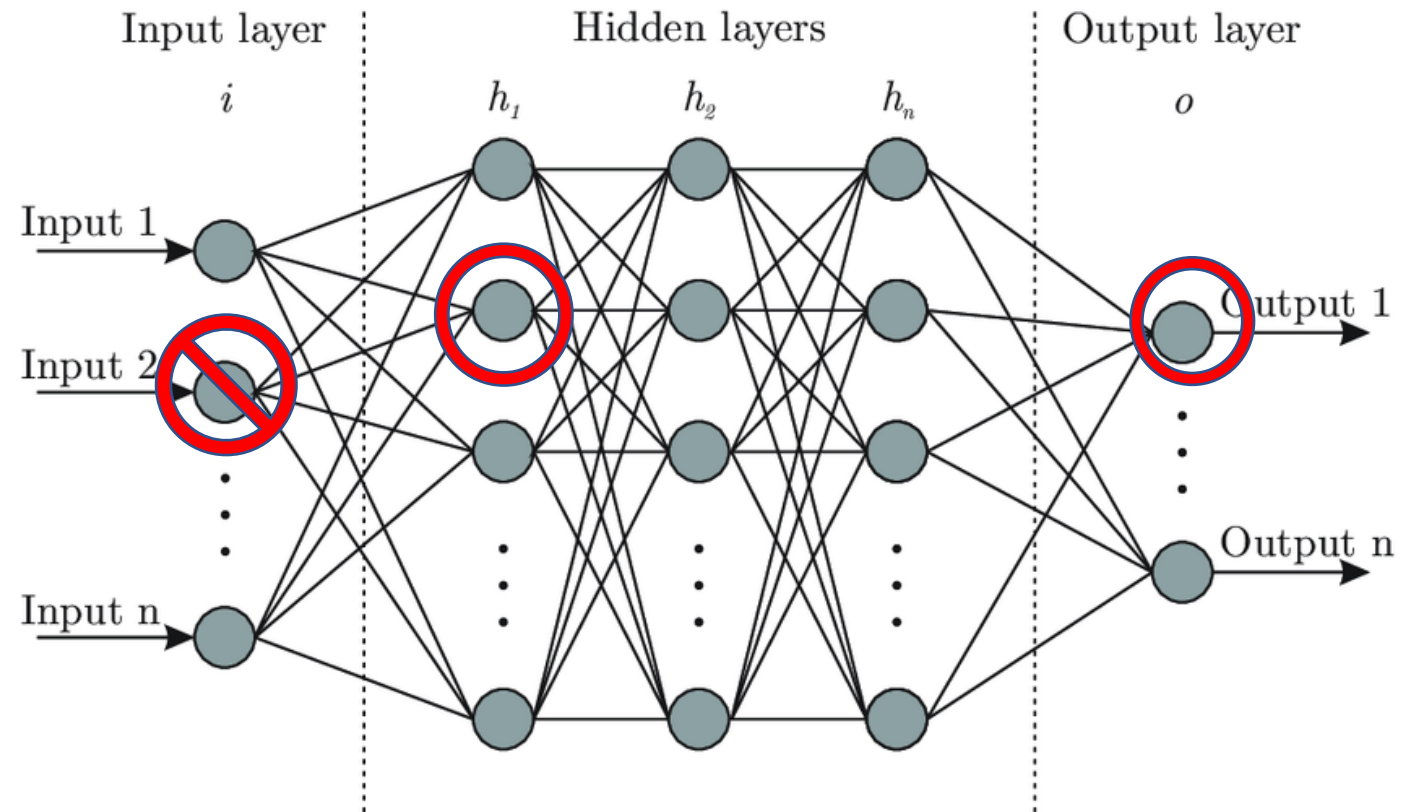
- Revisit perceptron
 - What is it? How does it work?
- Feed-forward neural networks
 - Basic architecture. How does it work?
- How does deep learning differ?
- Preview of future topics
 - Training (incl. backpropagation), CNNs, RNNs, GANs
- Tools
 - TensorFlow and Keras

Artificial Neural Net

Overview

Neural net overview

- ANNs composed of **units**
 - Neurons
 - Nodes in network
- Perceptron we saw earlier
- Inputs (several)
- Output (1, but copied)
- Function
 - Weights on inputs + bias
 - Activation function



The Unit (Perceptron)

One unit (neuron or cell) of an artificial neural network

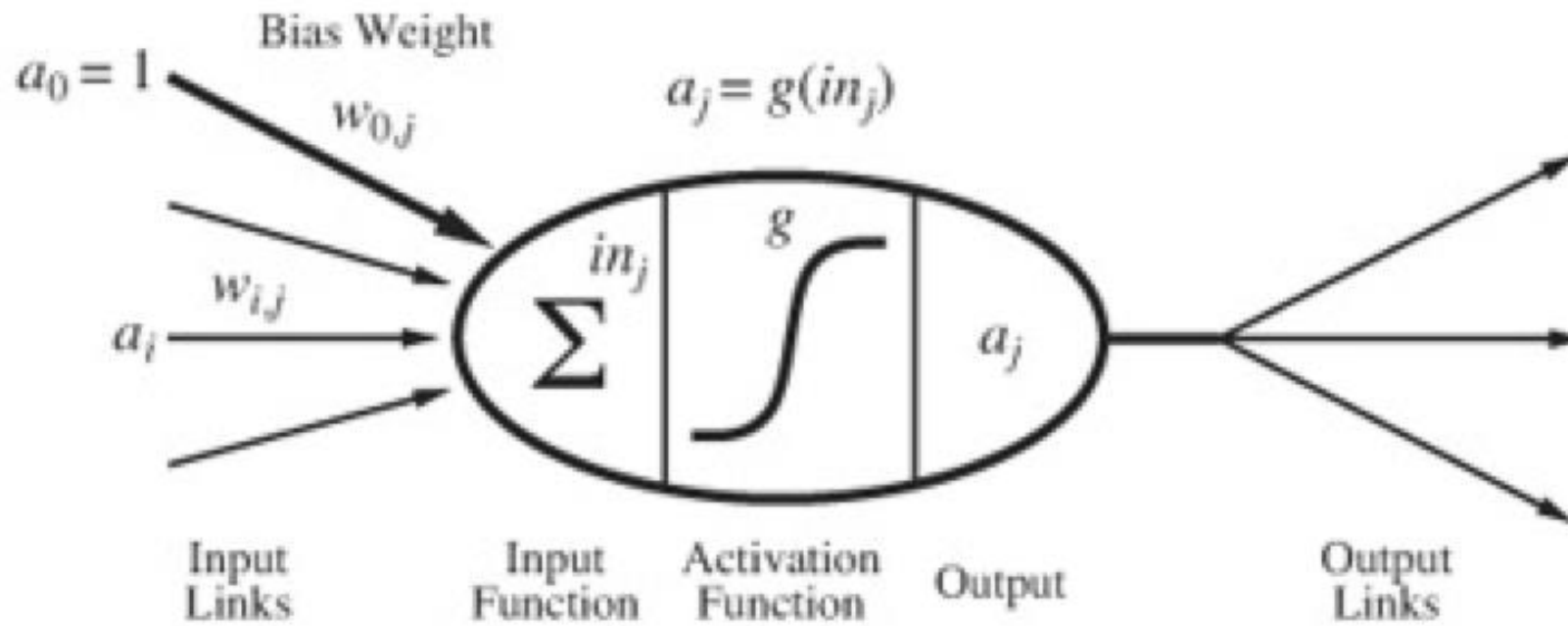
Unit (Perceptron)

- Inputs: **a** incl. $a_0 \equiv 1$
- Each feature multiplied by weight **w** incl. bias w_0
- Transform output with **activation function** (e.g., Heaviside, sigmoid):

$$g \left(\sum_{j=0}^m a_j \cdot w_j \right)$$

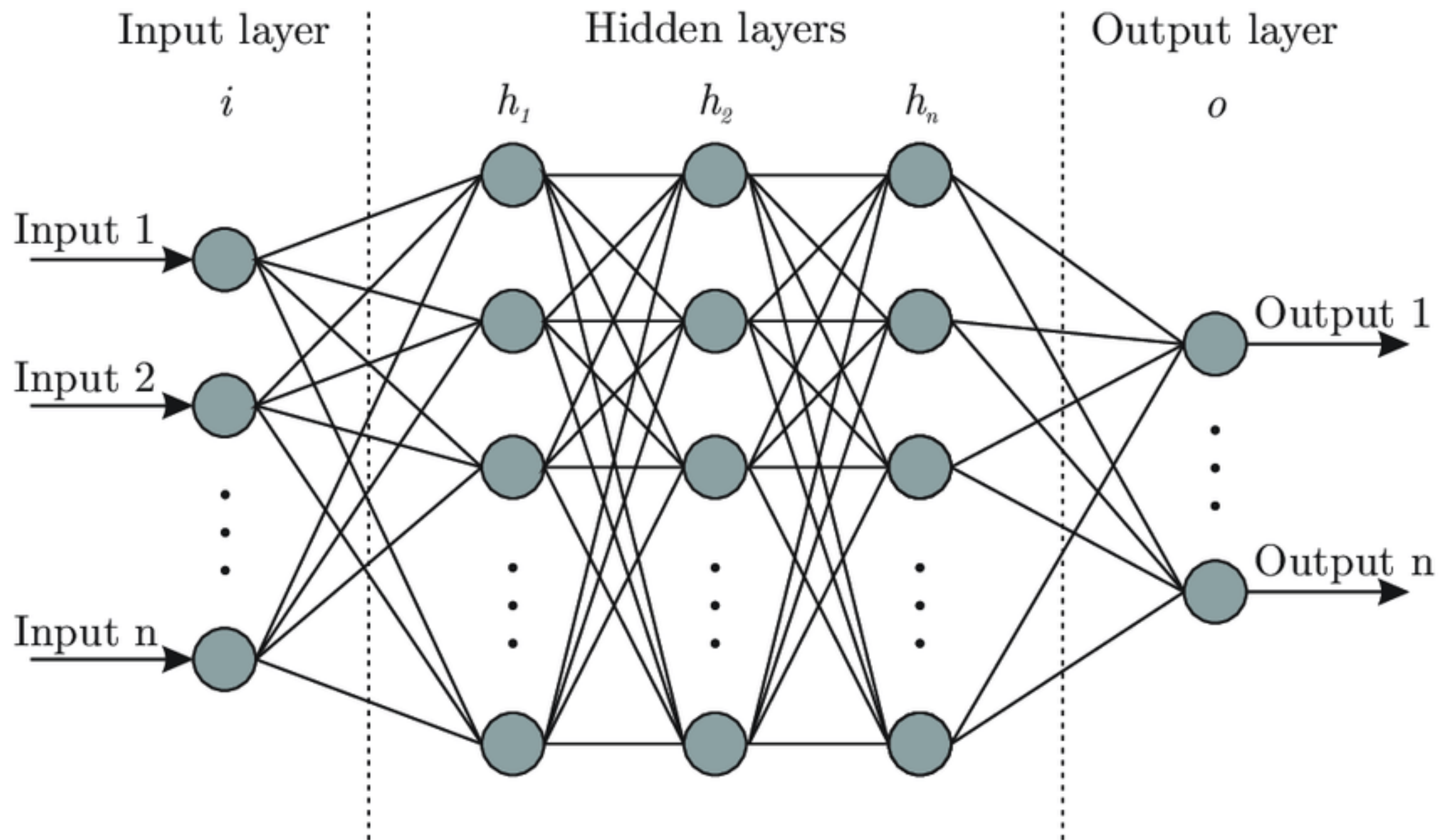
- Perceptron learning algorithm (originally hand-tuned)
- Linear model

Unit: How it works



Unit: How it works

- **Multiply** each **input** by its corresponding **weight**
- Take the **sum**
- Also add the **bias**
- Apply an **activation function** (many choices) to the sum
- **Output** the result
 - If **output layer** unit, then this is your **class label** or **predicted value**
 - If **hidden layer**, then feed into units in the next layer (next slide ...)
 - Known as forward propagation



Activation Functions

"Does the neuron fire?"

What is an Activation Function?

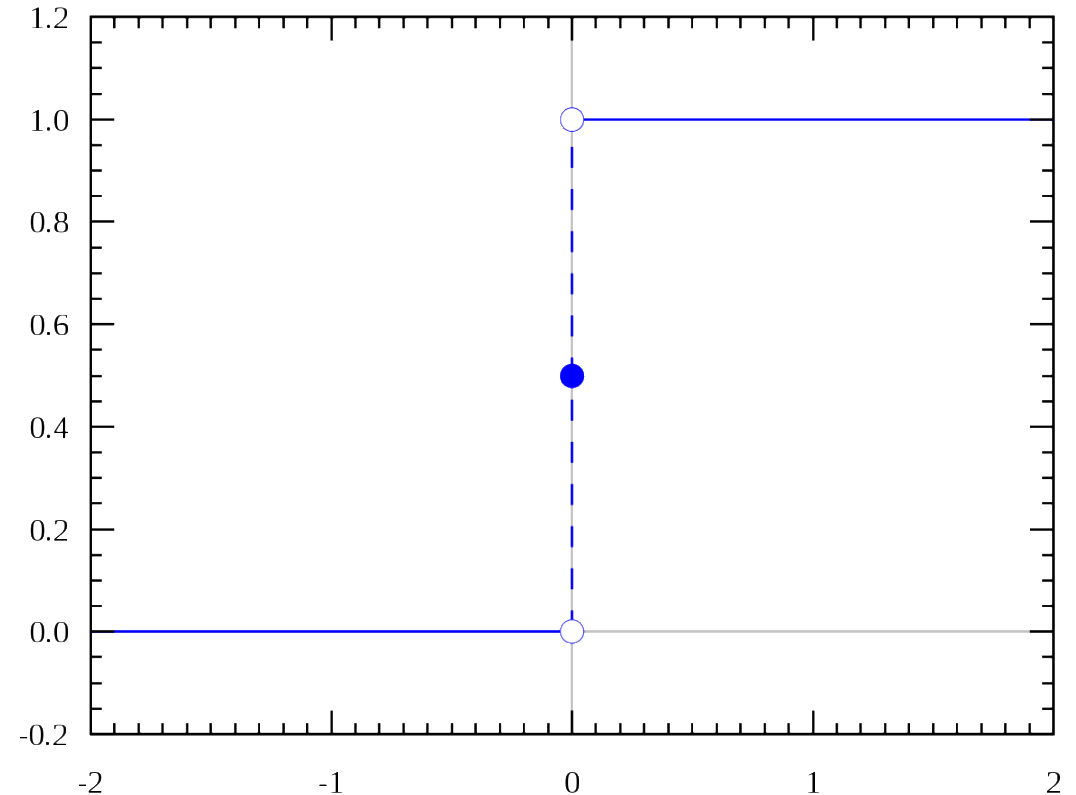
- Attached to each unit in network (except input layer)
- Applied after computing dot product

$$g \left(\sum_{j=0}^m x_j \cdot w_j \right)$$

- Determines whether the neuron "**fires**" (propagates forward)
- "**Is the input to this particular unit relevant to the model's prediction?**"
- Normalizes output to some value in a range, e.g., $[0, 1]$ or $[-1, 1]$
- Many choices! **Important hyperparameter!**

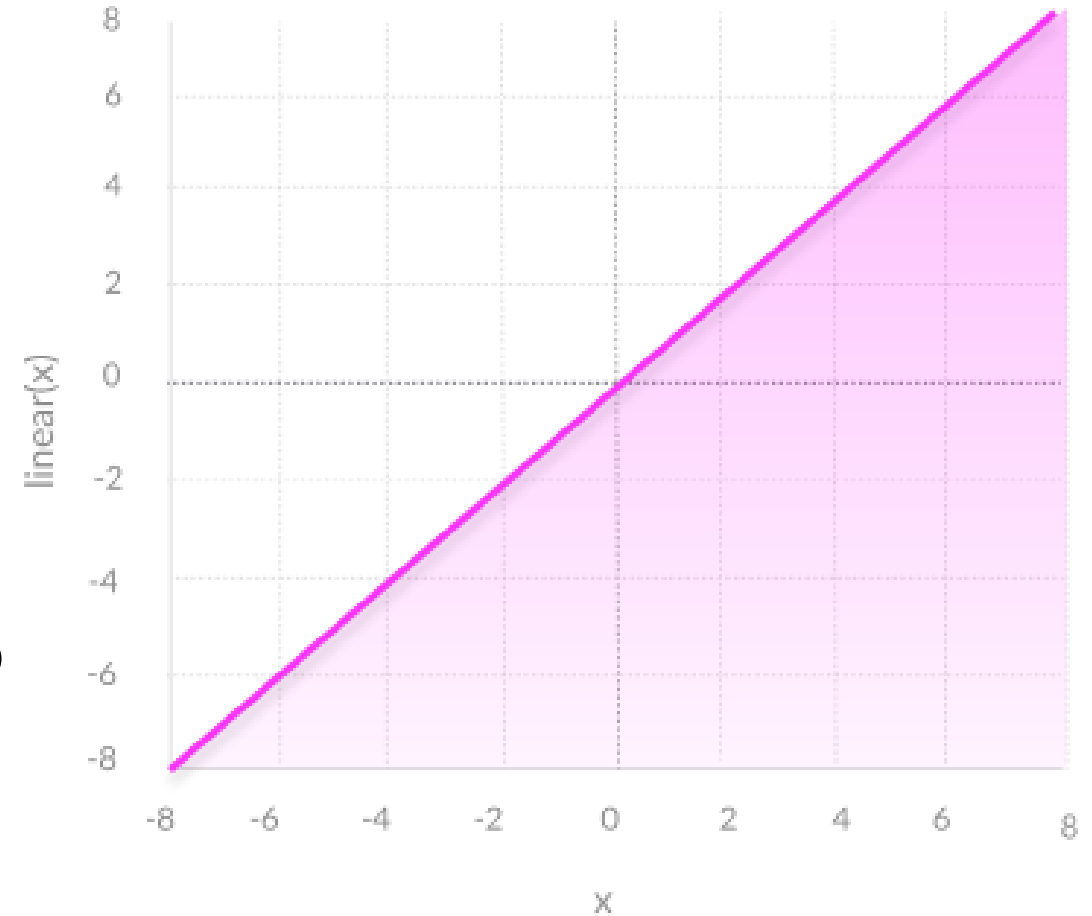
Choices for Activation Functions

- Heaviside (binary) step function
 - Not differentiable at $z = 0$
 - Everywhere else, derivative is 0
 - Backpropagation needs derivative
 - So can't train in the usual way ☹️



Choices for Activation Functions

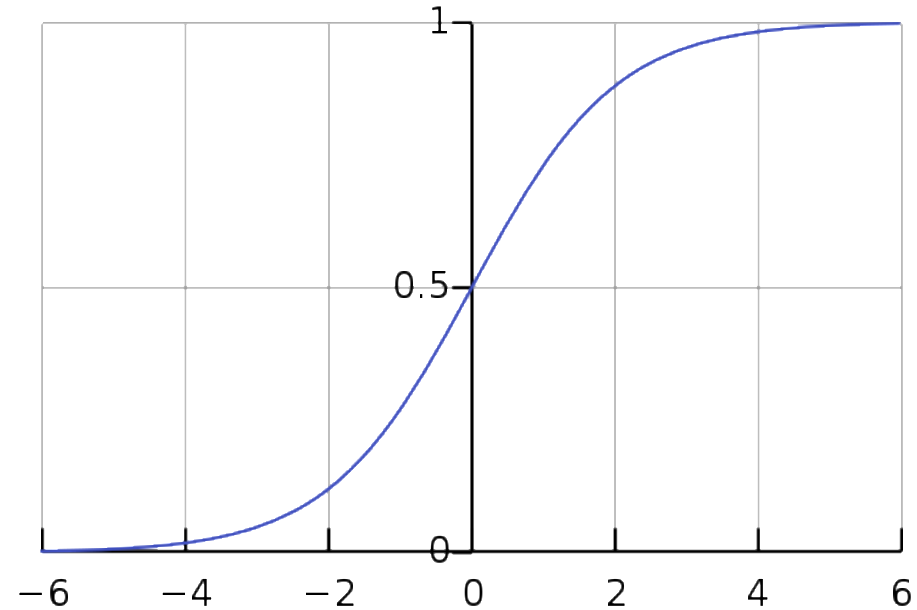
- Heaviside (binary) step function
- Linear function, e.g., $f(z) = z$
 - Note: No activation function is a special case of this
 - Derivative is a constant
 - So again, can't use backpropagation ☹️
 - Also, all layers collapse into one!
 - So we can only learn linear functions! ☹️



Choices for Activation Functions

- Heaviside (binary) step function
- Linear function, e.g., $f(z) = z$
- Sigmoid (logistic) function

$$f(z) = \frac{1}{1 + e^{-z}}$$



Advantages:

- Smooth gradient
- Output in range $[0,1]$
- Clear predictions

Disadvantages:

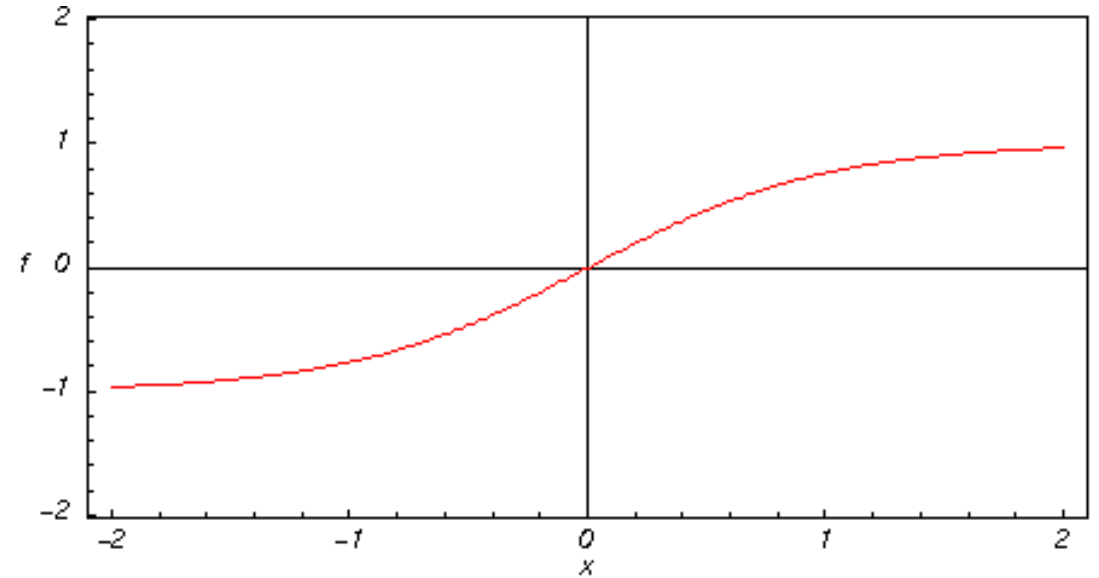
- Vanishing gradient (not enough info)
- Outputs not 0 centered
- Computationally expensive

Choices for Activation Functions

- Heaviside (binary) step function
- Linear function
- Sigmoid (logistic) function
- Tanh (hyperbolic tangent)

Advantages:

- Zero centered
- Otherwise like sigmoid

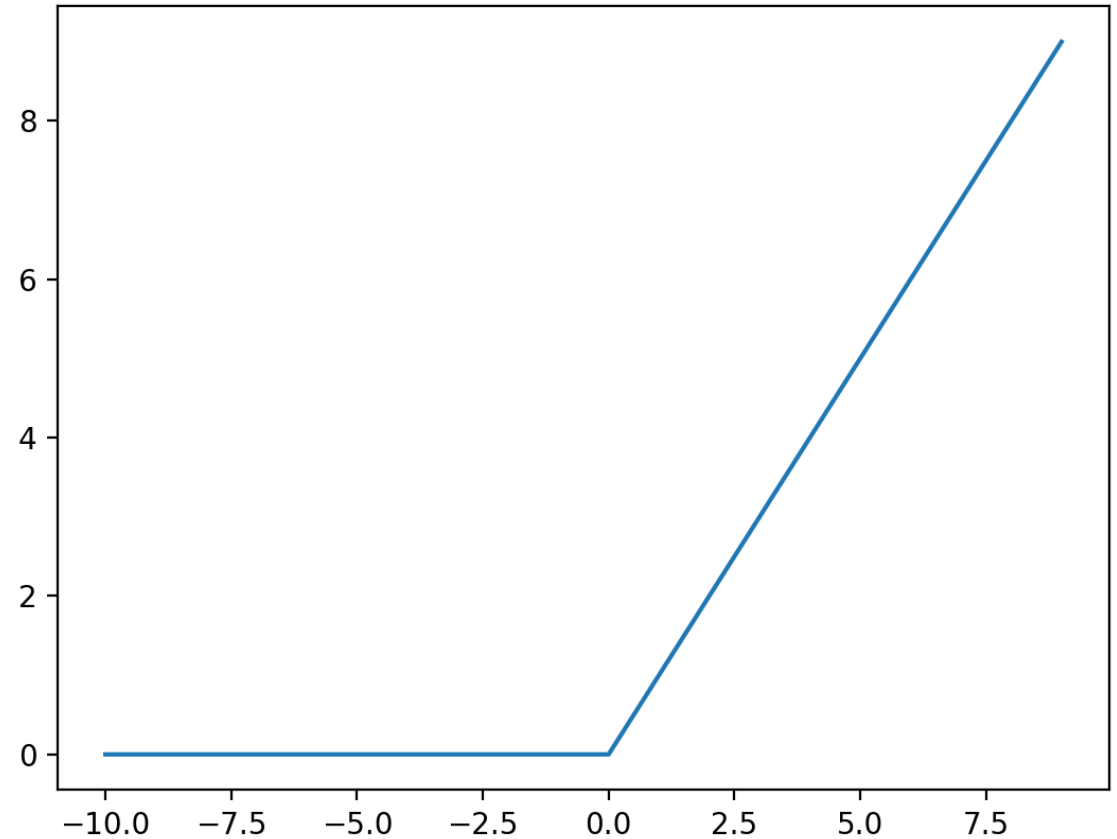


Disadvantages:

- Same as sigmoid

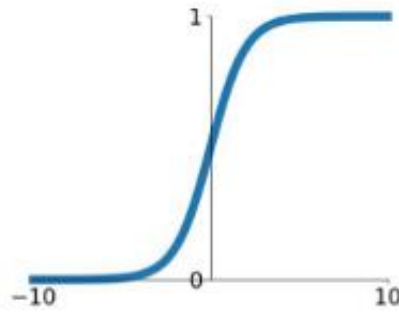
Choices for Activation Functions

- Heaviside (binary) step function
- Linear function
- Sigmoid (logistic) function
- Tanh (hyperbolic tangent)
- ReLU (rectified linear unit)
 - Computationally efficient
 - Non-linear, works with backprop.
 - But "dying ReLU problem" ☹️
- Today there are many choices ...



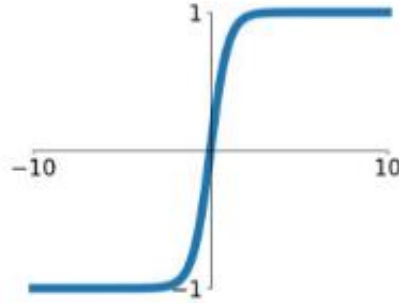
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



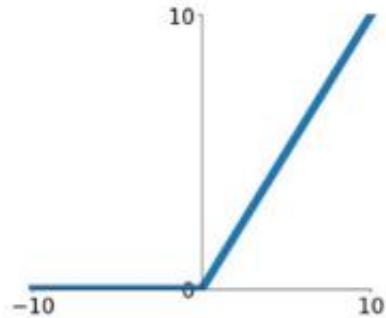
tanh

$$\tanh(x)$$



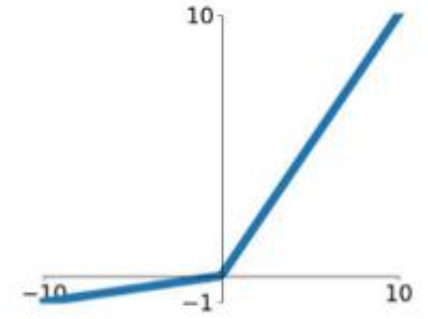
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

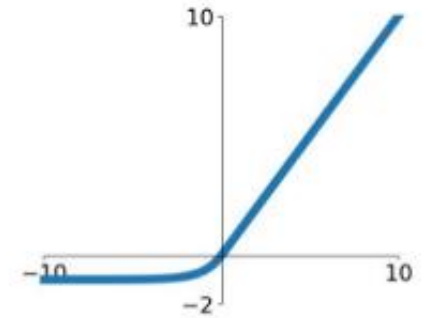


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$

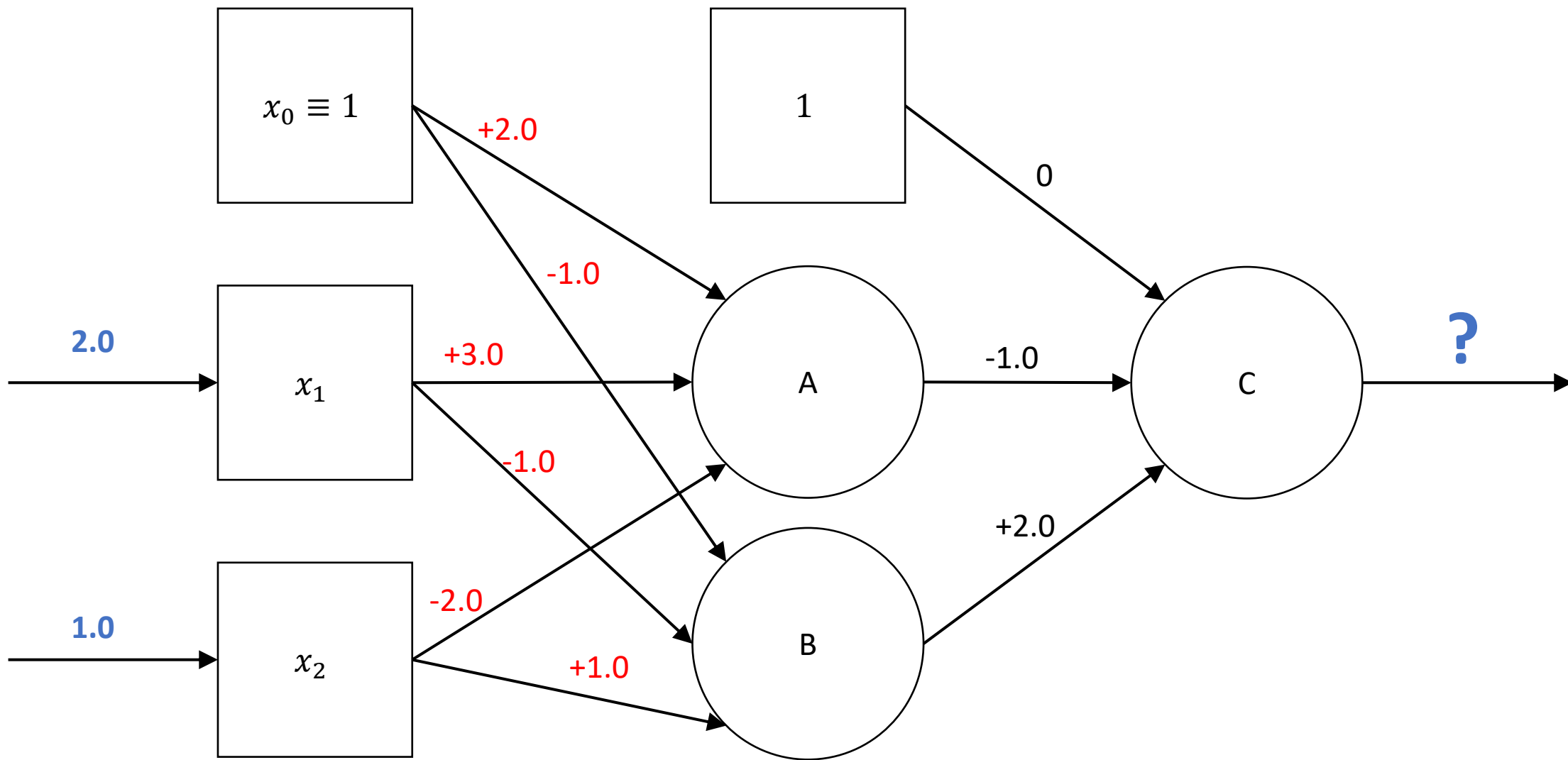


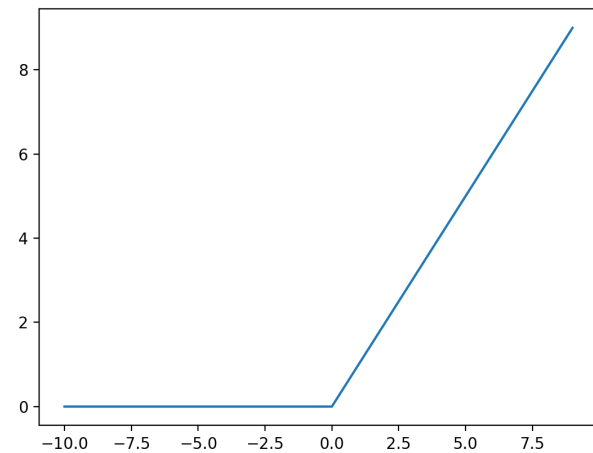
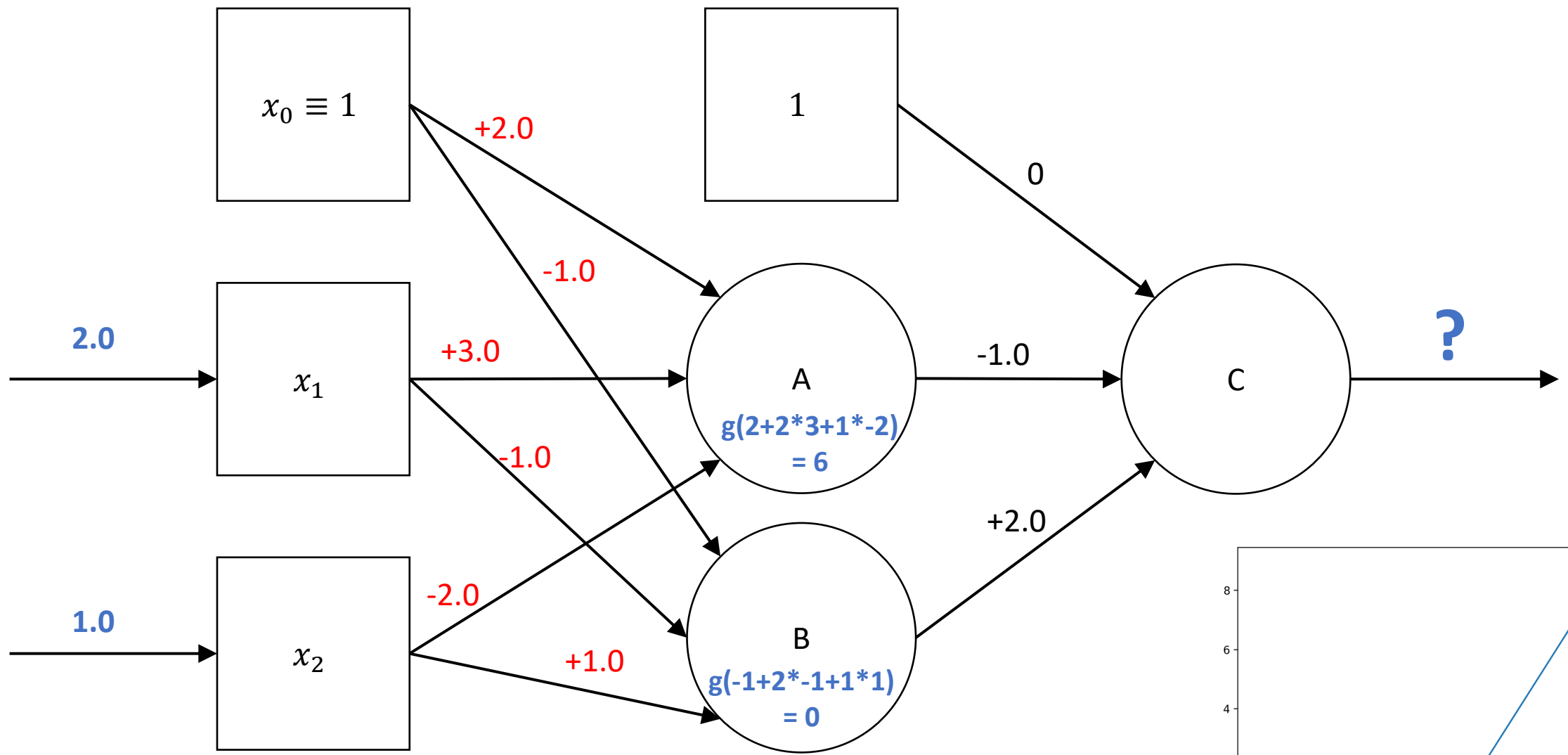
Feedforward Networks

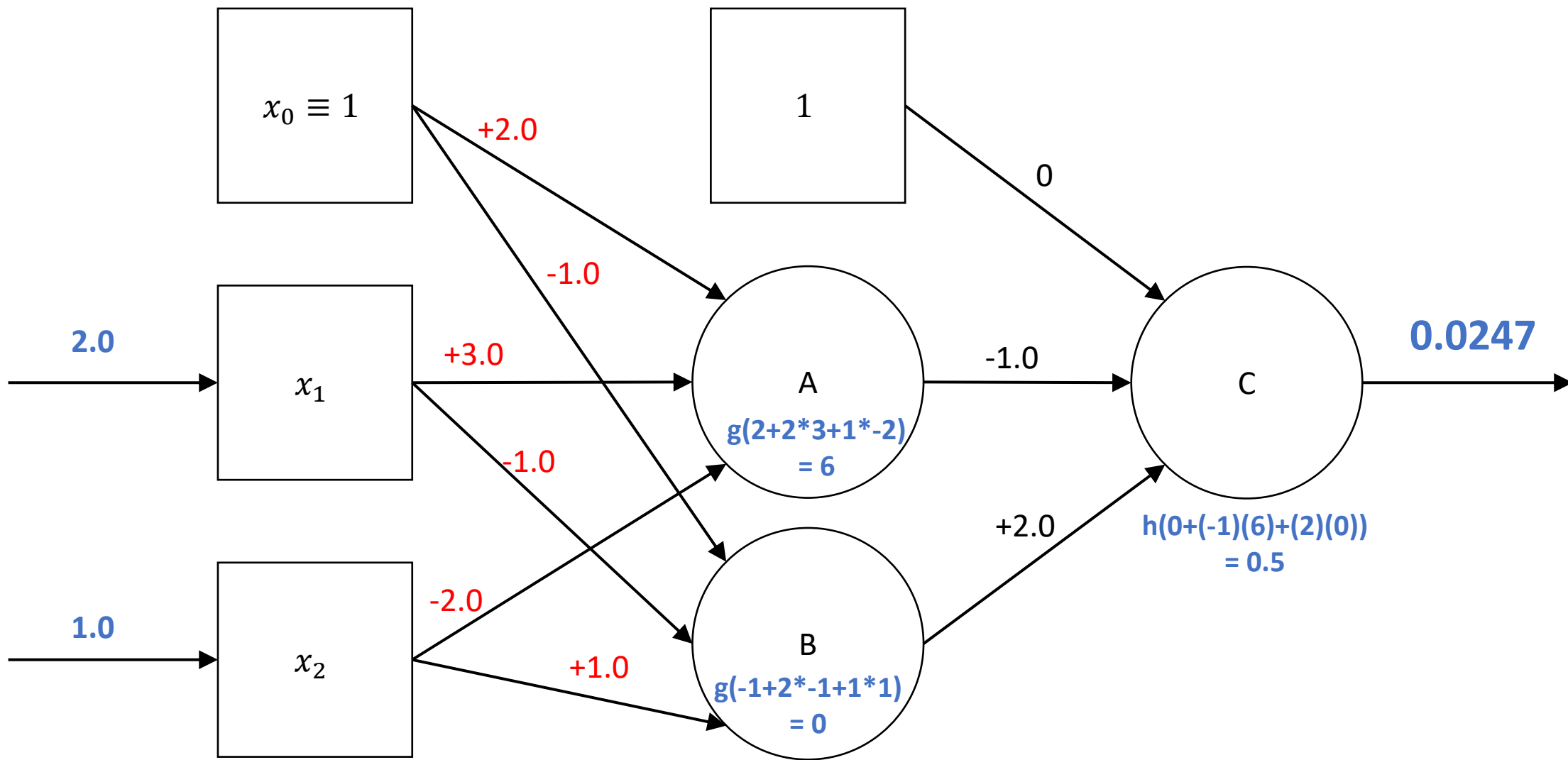
How do we use a neural network to make a prediction?

Using a Feedforward Network

- Suppose we have **learned** the network
- How do we use it to make a **prediction**?
- **Feed feature values** into the **input layer** (usually just $f(z) = z$)
- **Feed output of input layer** into **first hidden layer**
- Apply all units to compute unit output values
- **Feed** those outputs into the **next layer** (outputs become next inputs)
- Continue to feed forward through layer by layer
- Finally compute outputs of output layer (one neuron or many)







Feedforward Networks

- No cycles
 - Note: that would be a recurrent neural network (RNN)
- Output of each layer becomes the input to the next layer
- The last layer (output from output layer) gives us the results

What makes it a deep network?

- Deep learning is just a multi-layer perceptron
- **Usually with many hidden layers**
- Potentially complex structure
- Leverages new choices for activation function (e.g., ReLU)
- Computational resources: many times faster, memory, parallel (GPUs)

Parameters and Hyperparameters

- What are the parameters?
 - Weights and biases for all the units
- What are the hyperparameters?
 - Structure of the network (number hidden layers, number units, etc.)
 - Choice of activation function
 - There are many others (we will see them in future lessons ...)

Preview of Coming Attractions

What to expect the next few days

Preview

- How to **train** and **design** a network?
 - Backpropagation (finding the weights)
 - Hyperparameter selection
- Large Language Models (e.g., GPT-4o)

Tools

```
import tensorflow as tf
```

```
import tensorflow
```

```
import tensorflow as tf
```

Tools

- Don't use sklearn for real problems
- Go through tutorial (linked from Moodle):
<https://machinelearningmastery.com/tutorial-first-neural-network-python-keras/>
- Training is computationally expensive: GPUs very useful (buy or rent)

Recap

What did we learn?

What did we learn?

- What a neural network is
- How each unit (neuron) works
- Activation function, why important, options
- How to use a feedforward network to get a prediction
- Tools: tensorflow and keras in Jupyter notebook

What are your questions?

- Visit during office hours, or contact me via email, etc.!